

Loops

Learning Objectives

- Use repeat and forever loops
- Understand when to use count-controlled vs event-controlled loops
- Combine loops with other blocks for complex actions

Engage (5-7 minutes)

Think of grinding grain on a stone—you repeat the same circular motion over and over. Loops in Scratch do the same thing, repeating actions automatically so you do not have to write them again and again. Today we learn the power of repetition.

Explore (10-12 minutes)

Use “repeat 10” to make a sprite move in a pattern. Try “forever” to keep an action running endlessly. Combine loops with turns to create spirals and circles. Put a loop inside another loop and see what happens. Time how long a forever loop runs.

Explain (10-12 minutes)

Loops repeat blocks of code. “Repeat” runs actions a set number of times, while “forever” keeps running until you stop the program. Loops save time and make code cleaner. Nested loops put one loop inside another for complex repeating patterns, like a farmer planting rows of crops.

Elaborate (8-10 minutes)

Create a dance animation using loops—make the sprite move forward, turn, and repeat to create a pattern. Use “forever” to keep a character bouncing around the screen. Build a project where a loop changes costumes to create an animation, like frames of a traditional scroll painting.

Evaluate (5-8 minutes)

Exit ticket: What is the difference between “repeat” and “forever”? If you use “repeat 4” with “move 50 steps” and “turn 90 degrees,” what shape does the sprite draw? Why are loops useful in programming?